A bit of theory as far as relevant to the implementation of `class LagrangeMesh`

The necessary theory - for non-reduced axes - is found in Ryssens et al, PHYSICAL REVIEW C 92, 064318 (2015) *section III.B Lagrange-mesh representation*. The corresponding section in Baye, Phys. Rep. 565, 1 (2015) is *3.7.1. Lagrange–Fourier mesh*.

> **Note**
>
> The figures in this document are created by the tests. If the document complains `blablablah.png could not be found.`, please run the tests (`pytest tests` from folder `MOCCaPy`)

Non-reduced axes

(As from Ryssens et al, PHYSICAL REVIEW C 92, 064318 (2015) *section III.B Lagrange-mesh representation*)

Grid points

Using M evenly space grid point at distance Δ, the boundaries of the box are $[-\frac{M}{2}\Delta, \frac{M}{2}\Delta]$. The width of the box is $L = M\Delta$. We require $M = 2N$ to be even such that there are $N = \frac{M}{2}$ grid points on each side of the origin and the origin is avoided as a grid point.

For non-reduced axes (where the shift parameter σ is required be zero) the grid points are:
[eq 1]

$$\pm\frac{1}{2}\Delta, \pm\frac{3}{2}\Delta, \ldots, \pm\frac{2N-3}{2}\Delta, \pm\frac{2N-1}{2}\Delta$$

or
[eq 2]

$$x_{\pm i} = \pm(\frac{1}{2} + i)\Delta = \pm(\frac{2i+1}{2})\Delta 12$$

$$i = 0..N-1$$

(the notation $x_{\pm i}$ is not entirely accurate as it must distinguish between $+0$ and -0.)
In increasing order:
[eq 3]

$$-\frac{2N-1}{2}\Delta, -\frac{2N-3}{2}\Delta, \ldots, -\frac{3}{2}\Delta, -\frac{1}{2}\Delta, \frac{1}{2}\Delta, \frac{3}{2}\Delta, \ldots, \frac{2N-3}{2}\Delta, \frac{2N-1}{2}\Delta$$

or
[eq 4]

$$x_i = -\frac{2N-1-2i}{2}\Delta = (\frac{1}{2}+i-N)\Delta$$
$$i = 0..(2N-1)$$

or
[eq 4.1]

$$x_i = -\frac{2N-1-2i}{2}dx = (\frac{1}{2}+i-N)dx$$
$$i = -(N-1)..(N-1)$$

Basis functions

The basis functions are plane waves:
[eq 5]

$$\phi_k(x) = \frac{1}{\sqrt{L}}\exp(\frac{2\pi\mathrm{j}}{L}kx)$$

where we choose j for the imaginary unit (rather than j, as j and j are a bit better distinguishable than j and i), and

$$k = \pm\frac{1}{2}, \pm\frac{3}{2}, \ldots, \pm\frac{2N-3}{2}, \pm\frac{2N-1}{2}$$

or, since $k\Delta$ is a grid point, say, the i-th, x_i:
[eq 5]

$$\phi_{x_i}(x) = \frac{1}{\sqrt{L}}\exp(\frac{2\pi\mathrm{j}}{L}\frac{x_i}{\Delta}x) = \frac{1}{\sqrt{L}}\exp(\frac{2\pi\mathrm{j}}{L\Delta}x_i x)$$

> **Note**
>
> Since $\exp(\mathrm{j}z) = \cos z + \mathrm{j}\sin z$ the real component of the 1D basis function is symmetric and the imaginary component is skew-symmetric. Consequently, we can use them for testing interpolation with Lagrange functions (see below) also on reduced grids.

$$\mathrm{Re}(\phi_{x_i}(x)) = \frac{1}{\sqrt{L}}\cos(\frac{2\pi}{L\Delta}x_i x)$$
$$\mathrm{Im}(\phi_{x_i}(x)) = \frac{1}{\sqrt{L}}\sin(\frac{2\pi}{L\Delta}x_i x)$$

First order derivatives are:

$$\frac{d}{dx}\mathrm{Re}(\phi_{x_i}(x)) = -\frac{1}{\sqrt{L}}\frac{2\pi}{L\Delta}\sin(\frac{2\pi}{L\Delta}x_i x)$$

$$\mathrm{Im}(\phi_{x_i}(x)) = \frac{1}{\sqrt{L}}\frac{2\pi}{L\Delta}\cos(\frac{2\pi}{L\Delta}x_i x)$$

Interpolation

The Lagrange interpolation functions are:
[eq 6]

$$f_i(x) = \frac{1}{2N}\frac{\sin(\frac{\pi}{\Delta}(x-x_i))}{\sin(\frac{\pi}{\Delta}\frac{x-x_i}{2N})}$$

By construction, the Lagrange interpolation functions $f_i(x)$. have the property of being equal to 1 at the i-th mesh point, x_i, and 0 at all others:
[eq 7]

$$f_i(x_j) = \delta_{ij}$$

Note that when evaluating $f_i(x_i)$, both the numerator and the denominator are 0, and Python yields a `nan`. We need to invoke l'Hôpitals's rule to obtain the result.
[eq 7.1]

$$\lim_{x\to x_i}\frac{1}{2N}\frac{\sin(\frac{\pi}{\Delta}(x-x_i))}{\sin(\frac{\pi}{\Delta}\frac{x-x_i}{2N})}$$

$$= \frac{1}{2N}\lim_{x\to x_i}\frac{\sin'(\frac{\pi}{\Delta}(x-x_i))}{\sin'(\frac{\pi}{\Delta}\frac{x-x_i}{2N})}$$

$$= \frac{1}{2N}\lim_{x\to x_i}\frac{\frac{\pi}{\Delta}\cos(\frac{\pi}{\Delta}(x-x_i))}{\frac{\pi}{2N\Delta}\cos(\frac{\pi}{\Delta}\frac{x-x_i}{2N})}$$

$$= \frac{2N}{2N}\lim_{x\to x_i}\frac{cos(0)}{cos(0)} = 1$$

In order to avoid testing for $\frac{0}{0}$ it suffices to add $\varepsilon \approx 1\mathrm{e}-9$ to x_i to avoid the `nan` and yield something close to 1.

The arguments of the two sine functions in eq 6 are the same, apart from a factor $1/2N$:
[eq 8]

$$f_i(x) = \frac{1}{2N}\frac{\sin(A_i(x)))}{\sin(\frac{A_i(x)}{2N})}$$

$$A_i(x) = \frac{\pi}{\Delta}(x-x_i)$$

This can be exploited for efficiency in a computation.
An arbitrary function $h(x)$ taking the values $h_i = h(x_i)$ on the grid points can be interpolated on an arbitrary point $x \in [-N\Delta, N\Delta]$ as:
[eq 9]

$$h(x) = \sum_{i=0}^{2N-1} h(x_i) f_i(x)$$

This is essentially a dot product $\mathbf{f} \cdot \mathbf{h}$.

Derivatives

The 1st and 2nd order derivatives of the Lagrange interpolation functions are:
[eq 10.1]

$$D_{ji}^{(1)} = \left.\frac{df_i(x)}{dx}\right|_{x=x_j} = \begin{cases} (-1)^{i-j} \frac{\pi}{2N\Delta} \frac{1}{sin(\pi(i-j)/2N)}, & \text{for } i \neq j. \\ 0, & \text{for } i = j. \end{cases}$$

[eq 10.2]

$$D_{ji}^{(2)} = \left.\frac{d^2 f_i(x)}{dx^2}\right|_{x=x_j} = \begin{cases} (-1)^{i-j+1} 2 (\frac{\pi}{2N\Delta})^2 \frac{cos(\pi(i-j)/2N)}{sin^2(\pi(i-j)/2N)}, & \text{for } i \neq j. \\ -\frac{\pi^2}{3\Delta^2}(1 - \frac{1}{(2N)^2}), & \text{for } i = j. \end{cases}$$

> **⚠ Warning**
>
> I suspect that there is a subtle problem with these formulas. When coding them the unit test failed giving the right values but the opposite sign. While attempting to derive the formulas i discovered that using the definition in eq 6, taking derivatives and evaluating in x_j leads to $(x_j - x_i)$ in the arguments of the sine functions, which is equal to $(j-i)\Delta$, and NOT $(i-j)$ as written in the formulas. The derivation proceeds by considering separate cases for odd and even $(j-i)$ and noting that $\cos \pi(2n)$ and $\sin \pi(2n+1)$ vanish, yielding the sign factor in eq 10.1 and 10.2., $(-1)^{(i-j)} = (-1)^{(j-i)}$, but obviously the change from $(i-j)$ to $(j-i)$ in the sign functions yields a sign flip.
> (suspicion confirmed)

By applying these to the expansion $\phi(x) = \sum_{i=0}^{2N-1} \phi(x_i) f_i(x)$ we obtain
[eq 11]

$$\frac{d\mathbf{h}}{dx} = \left.\frac{dh(x)}{dx}\right|_{x=x_j} = \sum_{i=0}^{2N-1} \left.\frac{df_i(x)}{dx}\right|_{x=x_j} h(x_i) = \sum_{i=0}^{2N-1} D_{ji}^{(1)} h(x_i) = \mathbf{D}^{(1)}\mathbf{h}$$

So, the column vector $\frac{d\mathbf{h}}{dx}$ of the derivatives of $\mathbf{h}$ at all grid points is found as a matrix product of $\mathbf{D}^{(1)}$ with the column vector of the values of $\mathbf{h}$ at all the gridpoints. As the 2nd derivative forms a matrix as well, we also have
[eq 12]

$$\frac{d^2\mathbf{h}}{dx^2} = \mathbf{D}^{(2)}\mathbf{h}$$

Note, that single differentiation toggles the symmetry behavior of the function: if $h(x)$ is a

symmetric function, $h'(x)$ is skew-symmetric, and *vice versa*. Consequentially, double differentiation toggles it twice, hence keeps the symmetry behaviour the same.

Reduced axes

Grid points

On reduced axes only the N grid points to the right of the origin are kept:
[eq 13]

$$\frac{1}{2}dx, \frac{3}{2}dx, \ldots, \frac{2N-3}{2}dx, \frac{2N-1}{2}dx$$

or
[eq 14]

$$x_i = (i + \frac{1}{2})dx$$

$$i = 0..N-1$$

(In de case of reduced axis a shift $\sigma < dx$ may be subtracted from each grid point.)

Interpolation

According to (eq 9) an arbitrary function $h(x)$ can be interpolated as:

$$h(x) = \sum_{i=0}^{2N-1} h(x_i) f_i(x)$$

If the x-axis is reduced, i.e. we require $h(x)$ to be symmetric ($h(-x) = h(x)$) or skew-symmetric ($h(-x) = -h(x)$), then the above equation becomes:

$$h(x) = \sum_{i=0}^{N-1} h(x_i) f_i(x) + \sum_{i=0}^{N-1} h(-x_i) f_{-i}(x)$$

where $f_{-i}(x)$ is the Lagrange interpolation function corresponding to the i-th grid point to the left of the origin
[eq 14.1]

$$f_{-i}(x) = \frac{1}{2N} \frac{\sin(\frac{\pi}{\Delta}(x - x_{-i}))}{\sin(\frac{\pi}{\Delta}\frac{x - x_{-i}}{2N})} = \frac{1}{2N} \frac{\sin(\frac{\pi}{\Delta}(x + x_i))}{\sin(\frac{\pi}{\Delta}\frac{x + x_i}{2N})}$$

(where again we must distinguish a bit sloppily between $i = +0$ and $i = -0$ as the first mesh point index right, resp. left of the origin).
This then becomes:

$$h(x) = \sum_{i=0}^{N-1} \begin{cases} h(x_i)(f_i(x) + f_{-i}(x)), & h \text{ is symmetric} \\ h(x_i)(f_i(x) - f_{-i}(x)), & h \text{ is skew-symmetric} \end{cases}$$

or

[eq 15]

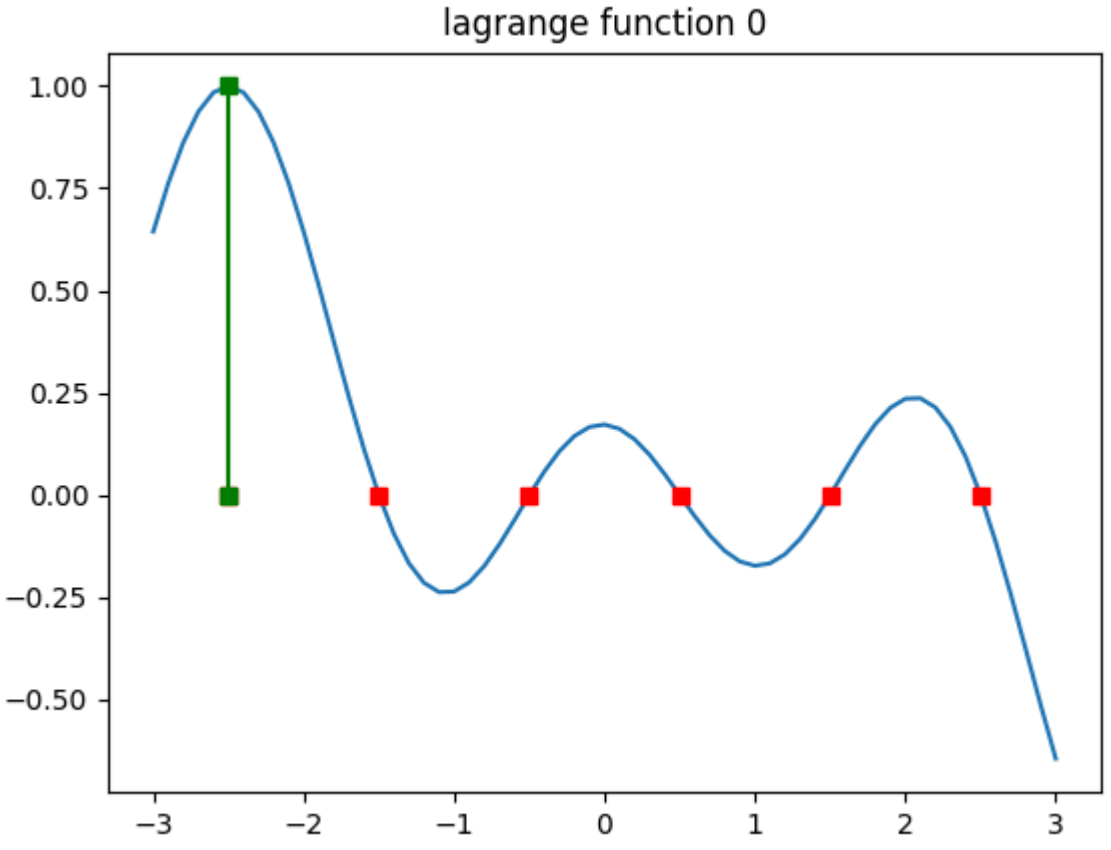
$$h(x) = \sum_{i=0}^{N-1} h(x_i)(f_i(x) \pm f_{-i}(x)) = \mathbf{h} \cdot \mathbf{f}_{\pm}$$

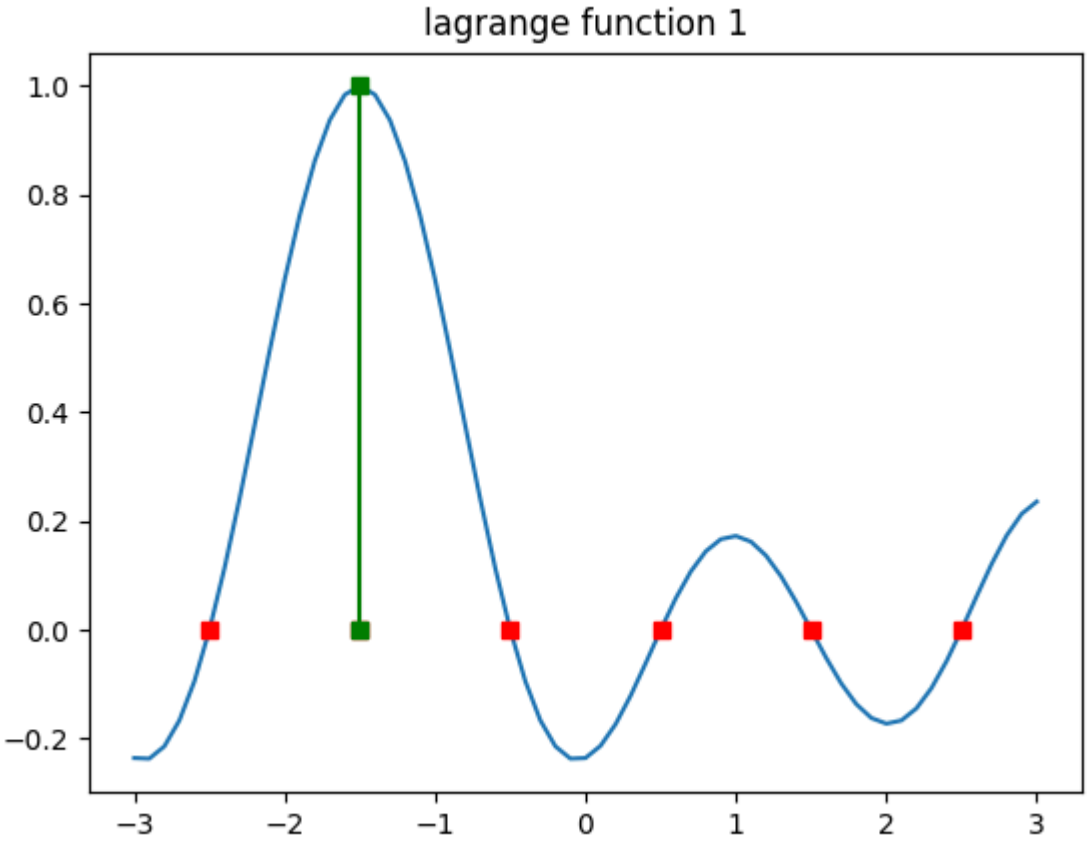
where the $\pm$ is $+$ for the symmetric case and $-$ for the skew-symmetric case, and $\mathbf{f}_{\pm} = \mathbf{f}_{+} \pm \mathbf{f}_{-}$

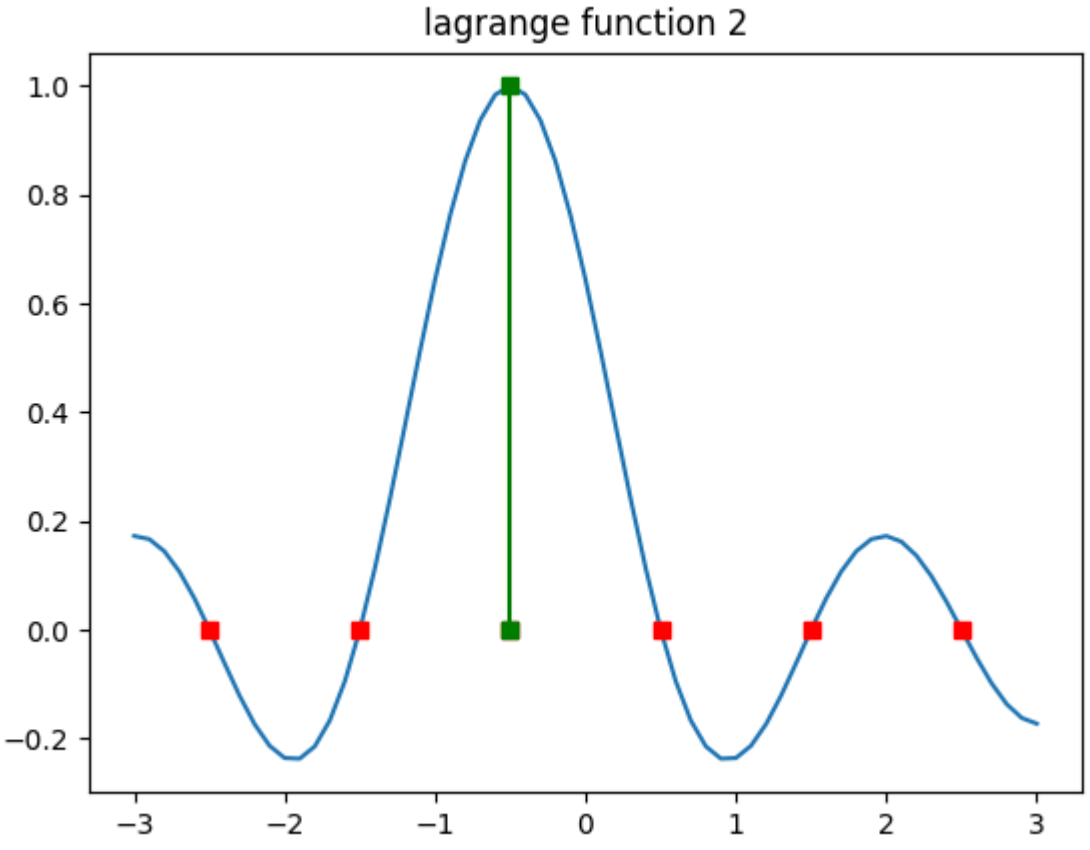
.

General remark on interpolation with Lagrange functions

On a 1D grid with 6 grid points at $-1.25, -.75, -.25, .25, .75, 1.25$ on the interval $[-1.5, 1.5]$, these are the 6 Lagrange functions:




lagrange function 1
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−3
−2
−1
0
1
2
3


lagrange function 2
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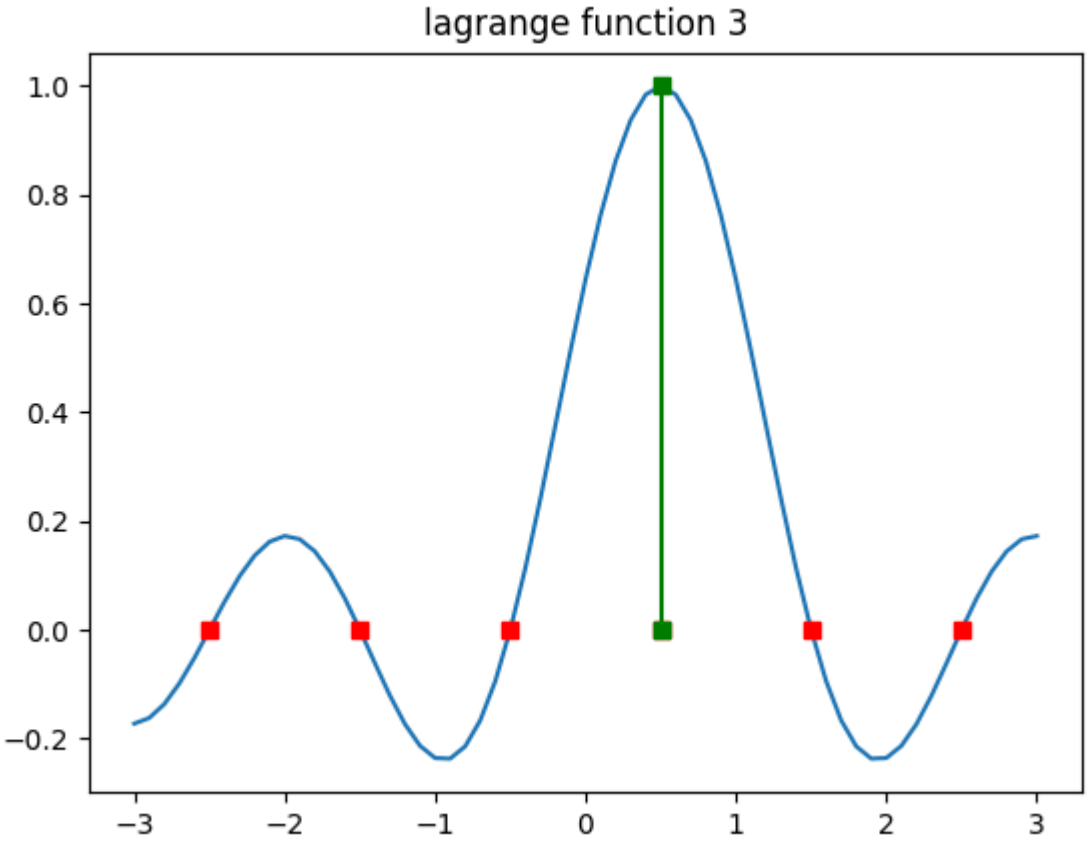
0.8
0.6
0.4
0.2
0.0
−0.2
−3
−2
−1
0
1
2
3


lagrange function 3
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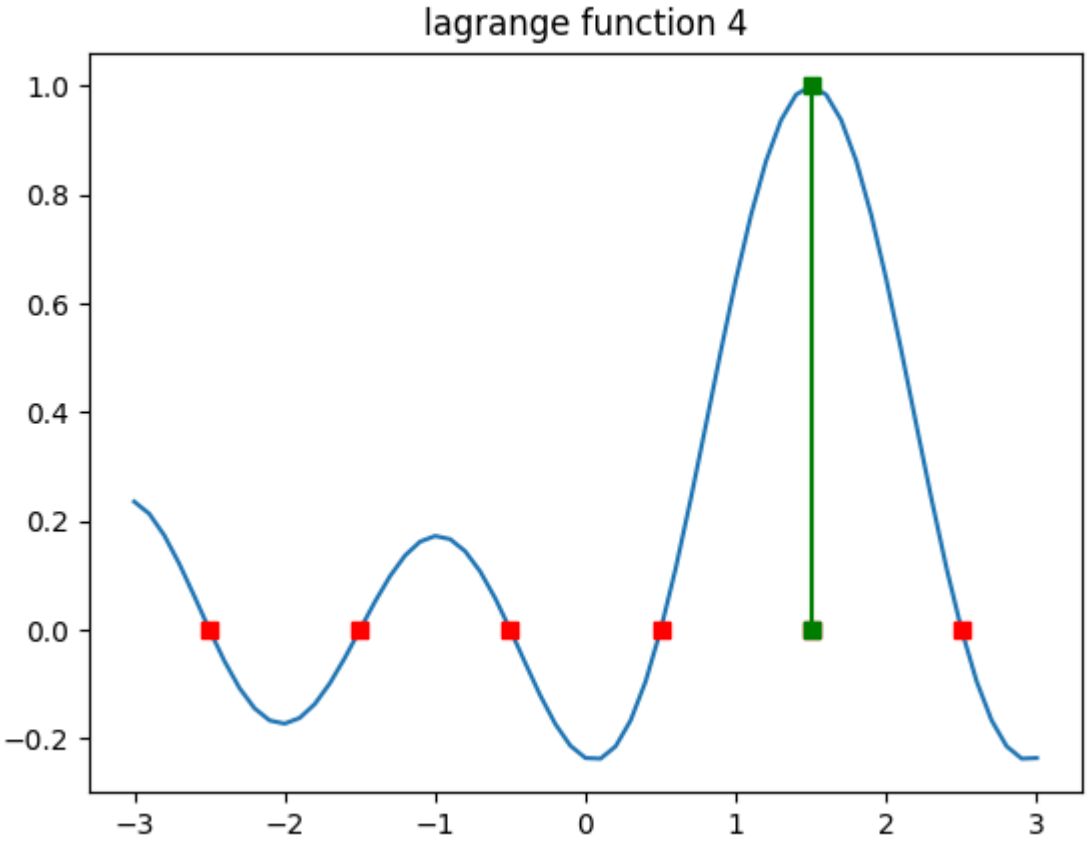
0.8
0.6
0.4
0.2
0.0
−0.2
−3
−2
−1
0
1
2
3


lagrange function 4
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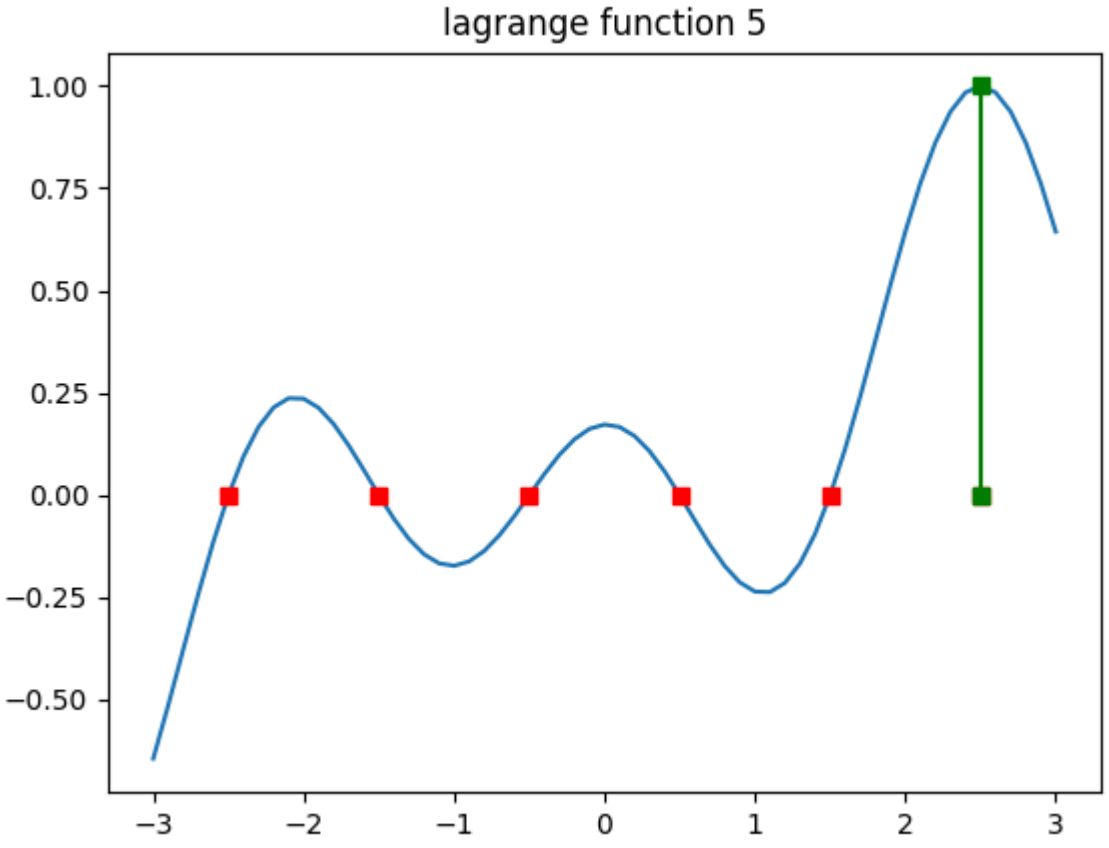
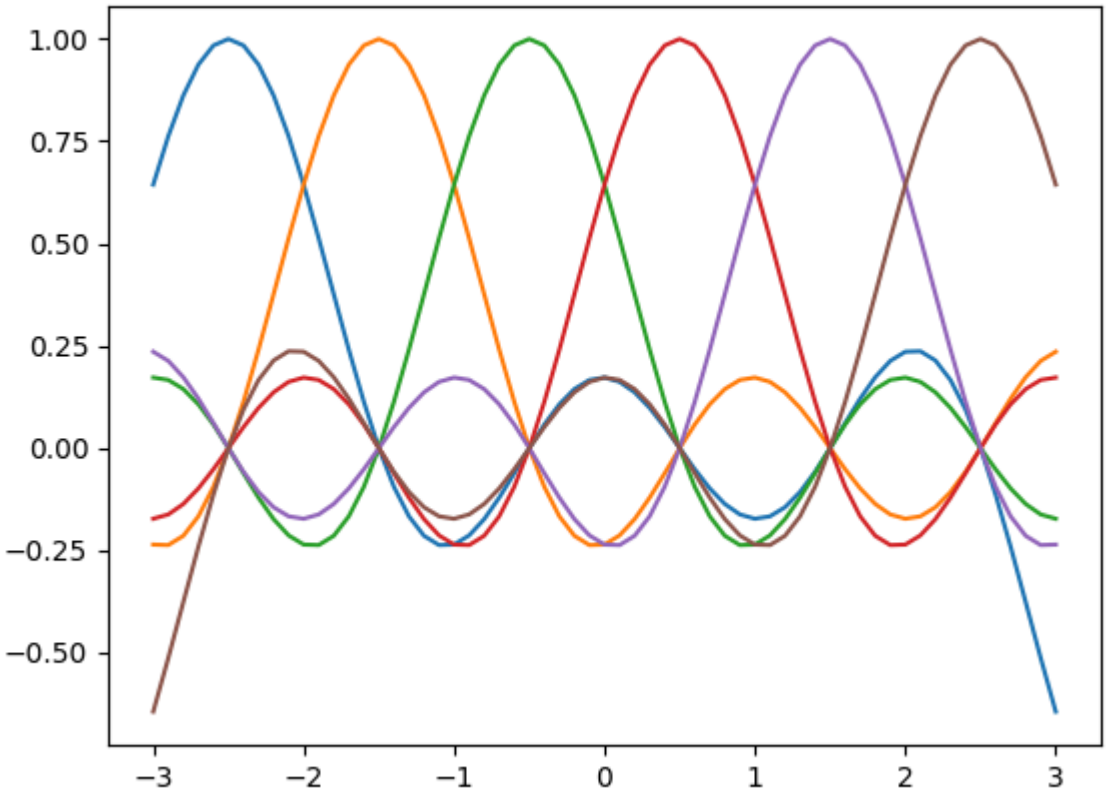
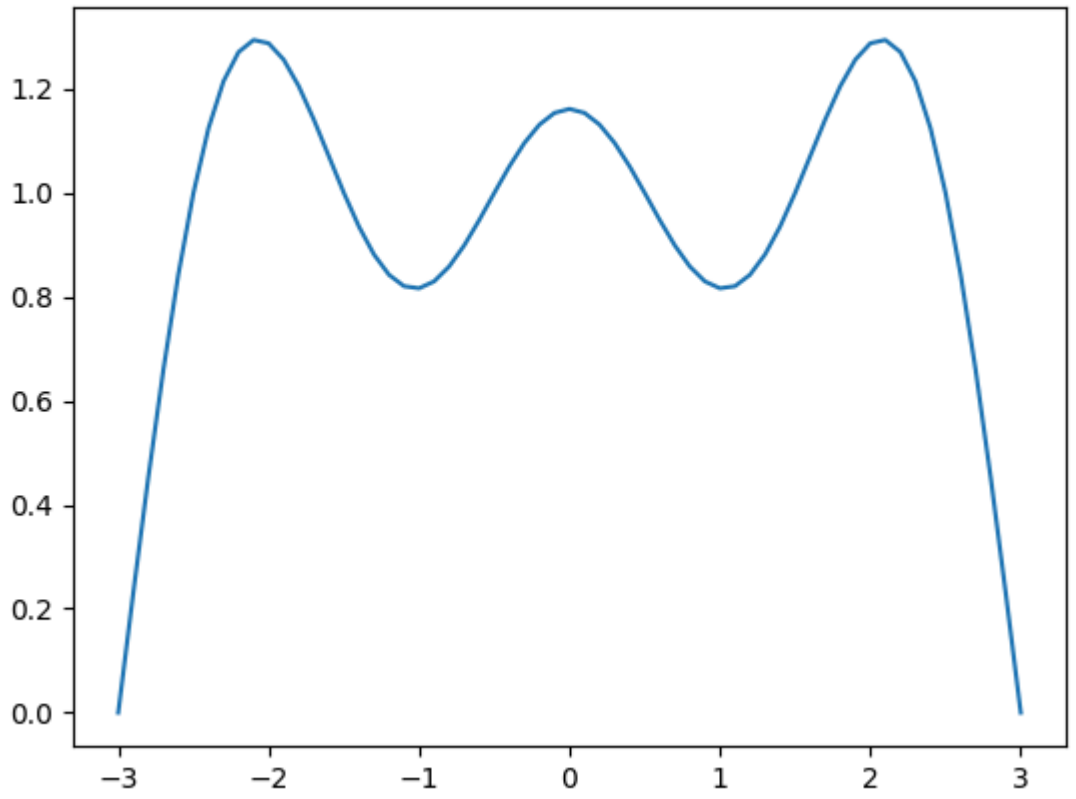
0.8
0.6
0.4
0.2
0.0
−0.2
−3
−2
−1
0
1
2
3





Clearly, each one yields 1 at one grid point and 0 at the others. Note also that they are antiperiodic: they reenter the box at the opposite edge, but **with a sign change**. Consequently, a constant function cannot be interpolated because it is periodic. The sum of

the 6 Lagrange functions is shown below. it is definitely not the constant function $f(x) = 1$.



According to the discussion in github issue 52 periodic functions can be interpolated with Lagrange functions provided they vanish at the boundary of the interval.

Derivatives

The formula for the derivative of a function h expanded on a reduced grid is found easily by extending the column vector $\mathbf{h}$ (of length N) on the reduced grid as
[eq 17]

$$\begin{bmatrix} \pm\mathbf{g} \\ -- \\ \mathbf{h} \end{bmatrix}$$

with $\mathbf{g}$ containing the same elements as $\mathbf{h}$ but in reverse order, as if they were mirrored by a horizontal line between the two. The full vector then corresponds to the $\mathbf{h}_{\mathrm{non-reduced}}$ case. This allows for the matrix product :
[eq 18]

$$\mathbf{D}\begin{bmatrix} \pm\mathbf{g} \\ \mathbf{h} \end{bmatrix} = \begin{bmatrix} \mathbf{D}^- \\ \mathbf{D}^+ \end{bmatrix}\begin{bmatrix} \pm\mathbf{g} \\ \mathbf{h} \end{bmatrix} = \begin{bmatrix} \mathbf{D}^-\begin{bmatrix} \pm\mathbf{g} \\ \mathbf{h} \end{bmatrix} \\ ---- \\ \mathbf{D}^+\begin{bmatrix} \pm\mathbf{g} \\ \mathbf{h} \end{bmatrix} \end{bmatrix}$$

The upper half of $\mathbf{D}$, $\mathbf{D}^-$, produces the derivatives on the negative x-axis, and the lower half of $\mathbf{D}$, $\mathbf{D}^+$, produces the derivatives on the positive x-axis.

> **Note**
>
> It can be demonstrated that $\mathbf{D}^T = -\mathbf{D}$, but apart from that $\mathbf{D}^-$ and $\mathbf{D}^+$ are unrelated. Thus $\mathbf{D}$ cannot be reduced to a rectangular matrix that can be applied to a reduced $\mathbf{h}$. (E.g. the transpose of the lower half of $\mathbf{D}$ cannot recover its upper left quadrant.)

The obvious way to proceed with differentiation for reduced axes is to construct the full $\mathbf{D}$-matrix and then applying eqs 17 and 18.

> **Note**
>
> According to eqs 10.1 and 10.2 the $\mathbf{D}$-matrix depends on the mesh only through N (or $M = 2N$) and Δ, not on the coordinates of the individual mesh points. So, its computation need not distinguish between the reduced and non-reduced case.

Because of the symmetry properties of h and differentiation, we only need, in fact

$$\mathbf{D}^+ \begin{bmatrix} \pm\mathbf{g} \\ \mathbf{h} \end{bmatrix}$$

An interesting question, from the performance point of view, is wether we can compute this result without explicitly constructing

$$\mathbf{h}^\dagger = \begin{bmatrix} \pm\mathbf{g} \\ \mathbf{h} \end{bmatrix}$$

which requires copying $\mathbf{h}$ twice.
We have, with $i = 0, ..., 2N-1$:

$$\left[\mathbf{D}^+ \begin{bmatrix} \pm\mathbf{g} \\ \mathbf{h} \end{bmatrix}\right]_i = \sum_{l=0}^{l=2N-1} D^+_{il} h^\dagger_l$$

=\sum*{l=0}^{l=N-1}D*{il}^+h/^*textdagger*+*sum*{l=N}^{l=2N-1}D_{il}^+h_l^\textdagger

$$where the first sum acts on \$\pm\mathbf{g}\$ and the second sum acts on \$\mathbf{h}\$.$$

=\sum*{l=0}^{l=N-1}D*{il}^+(\pm h*{N-1-l})*
+*sum*{l=N}^{l=2N-1}D*{il}*^+*h*{l-N}

=\pm\sum*{l=0}^{l=N-1}D*{il}^+h*{N-1-l}*
+*sum*{l=0}^{l=N-1}D*{i,N+l}*^+*h*{l}

$$If we divide \$\mathbf{D}\$ in 4 \$N \times N\$ quadrants: \$\$\mathbf{D} = \begin{bmatrix} \mathbf{D}^{\mathrm{ul}} & \mathbf{D}^{\mathrm{ur}} \\ \mathbf{D}^{\mathrm{ll}} & \mathbf{D}^{\mathrm{lr}} \end{bmatrix}$$

$$\left[\mathbf{D}^{+}\begin{bmatrix}\pm\mathbf{g}\\ \mathbf{h}\end{bmatrix}\right]_i = \pm\sum_{l=0}^{l=N-1} D^{\mathrm{ll}}_{il} h_{N-1-l} + \sum_{l=0}^{l=N-1} D^{\mathrm{lr}}_{i,l} h_l$$

now with $i = 0, ..., N-1$.

> **Note**
>
> The second sum can be implemented using `numpy.einsum`, as in the non-reduced case. For the first sum, this is perhaps not possible because the access to h is reversed.

This expression does not necessitate the explicit construction of $\mathbf{h}^{\dagger}$, but complicates the implementation. At this point, it seems wise to keep the implementation as simple as possible, sacrificing - perhaps, this is not even sure - a bit of performance.

N-dimensional grids

The full Cartesian 3D representation of a function $h(\mathbf{r})$, where $\mathbf{r} = [x \quad y \quad z]$ (3D case), is then provided (for the 3D case) by
[eq 23]

$$h(\mathbf{r}) = \sum_{ijk} h_{ijk} f_i(x) f_j(y) f_k(z)$$

where the number of discretization points does not have to be the same in each direction.

Note that f_i, f_j and f_k are generally different objects, even if accidentally the indices i, j and k are identical, as they pertain, resp., to the x-axis, the y-axis and the z-axis, and the grid spacing is not necessarily the same.

Derivatives

In this case, the derivative matrices $\mathbf{D}^{(1)}$ and $\mathbf{D}^{(2)}$ have to be set up separately for each direction, taking into account wether the axis is reduced or not.
[eq 23.1]

$$\left.\frac{d\Phi}{dx}\right|_{ijk} = \sum_l D^1_{il}\Phi_{ljk}$$

$$\left.\frac{d\Phi}{dy}\right|_{ijk} = \sum_l D^1_{il}\Phi_{ilk}$$

$$\left.\frac{d\Phi}{dz}\right|_{ijk} = \sum_l D^1_{il}\Phi_{ijl}$$

The summation is over the index of Φ_{ijk} that corresponds to the axis wrt which the derivative is taken: the derivative wrt $x/y/z$ sums over the 1st/2nd/3rd index. The same holds for 2nd, 3rd, 4th order derivatives ($\frac{d^n}{du^n}$, $u = x, y, z$, $n \geq 0$).

> **Note**
>
> `numpy.einsum` is ideally suited to compute sums like these.

Basis functions

The basis functions are plane wave products of the different axes:
[eq 24]

$$\Phi_{klm}(x,y,z) = \phi_k(x)\phi_l(y)\phi_m(z) = \frac{1}{\sqrt{L_x}}\frac{1}{\sqrt{L_y}}\frac{1}{\sqrt{L_z}}\exp(\frac{2\pi \mathrm{j}}{L_x}kx)\exp(\frac{2\pi \mathrm{j}}{L_y}ly)\exp(\frac{2\pi \mathrm{j}}{L_z}mz)$$

$$= \frac{1}{\sqrt{L_xL_yL_z}}\exp(2\pi \mathrm{j}(\frac{kx}{L_x} + \frac{ly}{L_y} + \frac{mz}{L_z})$$

$$k = \pm\frac{1}{2}, \pm\frac{3}{2}, \ldots, \pm\frac{N_x - 1}{2}$$

$$l = \pm\frac{1}{2}, \pm\frac{3}{2}, \ldots, \pm\frac{N_y - 1}{2}$$

$$m = \pm\frac{1}{2}, \pm\frac{3}{2}, \ldots, \pm\frac{N_z - 1}{2}$$

$$= \frac{1}{\sqrt{L_xL_yL_z}}\exp(2\pi \mathrm{j}(\mathbf{k}\cdot\mathbf{r})$$

$$\mathbf{k} = \begin{bmatrix} \frac{k}{L_x} & \frac{l}{L_y} & \frac{m}{L_z} \end{bmatrix}$$

$$\mathbf{x} = \begin{bmatrix} x & y & z \end{bmatrix}$$

Note that ϕ_k, ϕ_l and ϕ_m are generally different objects, even if accidentally the indices k, l and m are identical, as they pertain, resp., to the x-axis, the y-axis and the z-axis.
Alternatively, in the spirit of eq 5.1:
[eq 24.1]

$$\mathbf{k} = \begin{bmatrix} \frac{x_i}{L_x\Delta_x} & \frac{y_j}{L_y\Delta_y} & \frac{z_k}{L_z\Delta_z} \end{bmatrix}$$

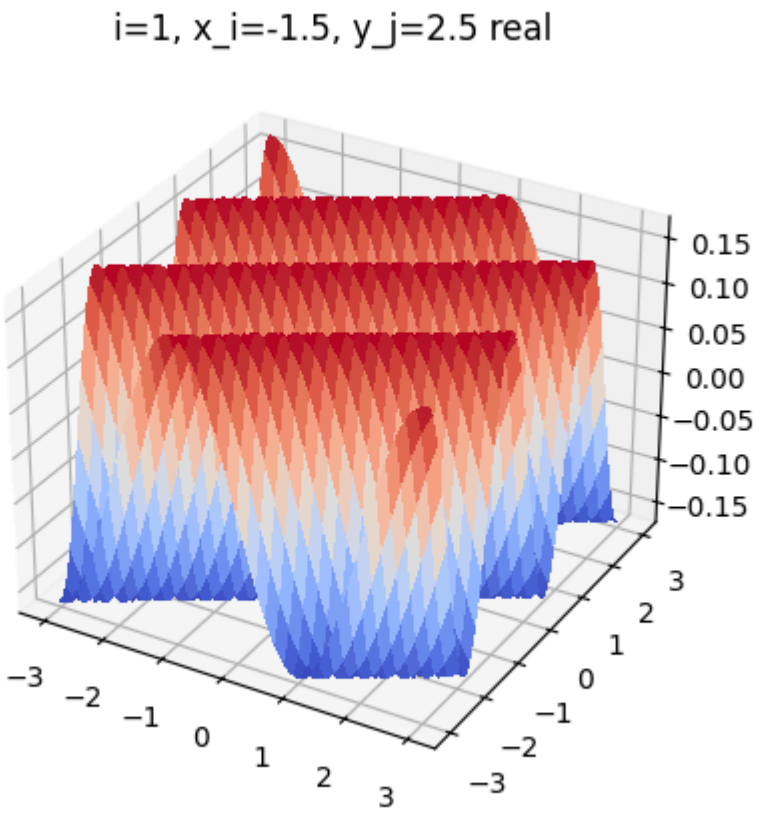
where the x_i, y_j, z_k are grid coordinates in the x, y and z directions.

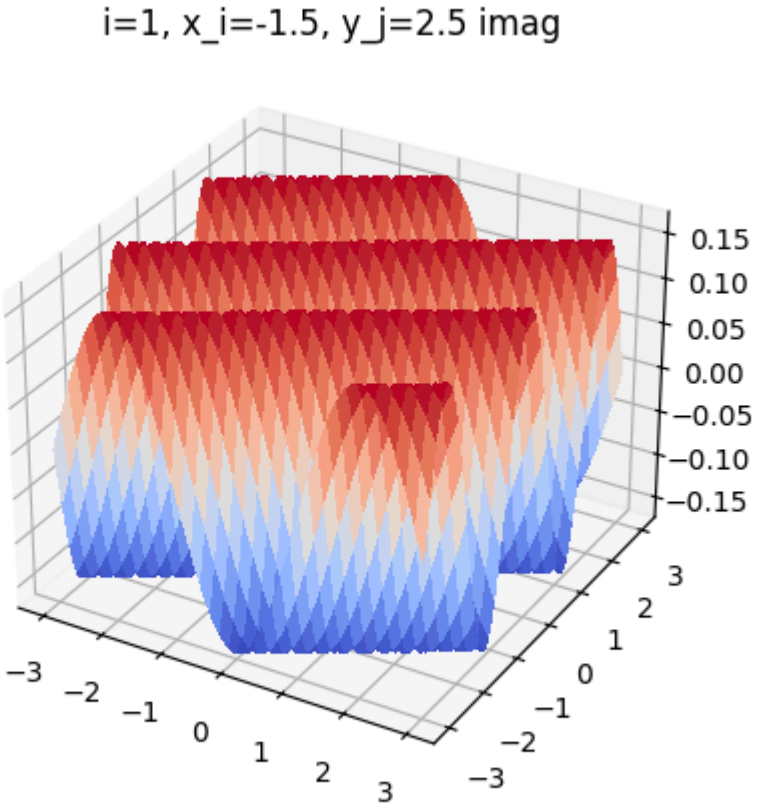
> **Note**
>
> Contrary to the 1D case, it is generally not true for the 2D and 3D cases that the basis functions are symmetric or skew-symmetric.

E.g. changing the sign of the x-coordinate in $\exp(2\pi \mathrm{j}(\mathbf{k} \cdot \mathbf{r}) = \exp(2\pi \mathrm{j}(\mathbf{k} \cdot [-x, y, z])$ yields

$$\cos(-k_x x + k_y y + k_z z) + \mathrm{j} \sin(-k_x x + k_y y + k_z z)$$

and $\cos(-k_x x + k_y y + k_z z) = \cos(k_x x + k_y y + k_z z)$ only if $k_y y + k_z z$ is a multiple of 2π, which is generally not true. This can also be seen in the figures below. The plane waves are periodic only in the direction of the wave vector.





Interpolation

As described by eq 23 Interpolating a scalar quantity h at a single point requires a sum over all grid points which may be costly (speaking of working interactively). If h needs to be interpolated on a large number of points, p,

$$\mathbf{r} = \begin{bmatrix} x_0 & y_0 & z_0 \\ x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ \vdots & \vdots & \vdots \\ x_{p-1} & y_{p-1} & z_{p-1} \end{bmatrix}$$

it will be advantageous to move the loop over the points $0..p-1$ inside the loop over ijk product $h_{ijk}f_i(x)f_j(y)f_k(z)$.
In case h is not a scalar quantity but a tensor it is advantageous to move the loop over the components between the loop over the p interpolation points and the loop over ijk:

```
for all grid points ijk:
    for all components h of H
        for all interpolation points 0..p
            accumulate h_ijk * f_i(x_p) * f_j(y_p) * f_k(z_p) over ijk
```

The inner loop can be evaluated as a function over a numpy array:

```
for all grid points ijk:
    for all components h of H
        h(r) +=  h_ijk * f_i(r[:,0]) * f_j(r[:,1]) * f_k(r[:,2])
```

Furthermore, in case e.g. the x axis is reduced, according to eq 15 one must replace $f_i(x)$ by $(f_i(x) \pm f_{-i}(x))$, where the sign is $+$ if h is symmetric w.r.t. x and $-$ if h is skew-symmetric w.r.t. x .